

Trainings ▾

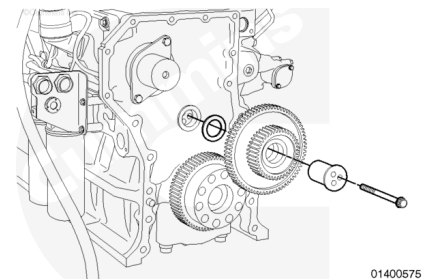
Preparatory Steps

- Remove the oil pan and associated components. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16 (</qs3/pubsys2/xml/en/procedures/89/89-016-006.html>).
- Remove the camshaft gear. Refer to Procedure 001-008 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-008.html>)

Remove

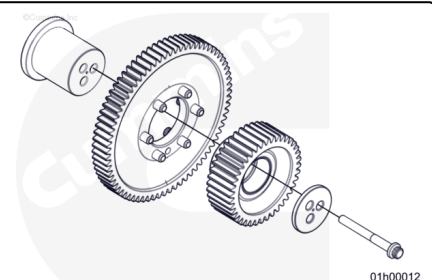
⚠ CAUTION ⚠

Check the punch mark on the mounting capscrew head. If seven marks are punched, replace the capscrew with a new capscrew. Failure to replace capscrew could result in breakage.



For CKEC engines built before engine serial number (ESN) 00327150, and PHP engines built after ESN 85000848, remove the three capscrews, the shaft, the camshaft idler gear assembly, and the rear thrust bearing.

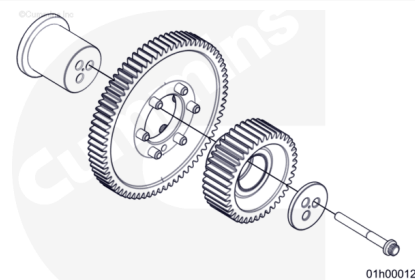
For CKEC engines built after ESN 00327150, and PHP engines built after ESN 85000848, remove the three capscrews, the shaft, the camshaft idler gear assembly, and the gear retainer.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use solvent. Clean the idler gear and parts.

Dry with compressed air.

Check parts for damage.

Measure the inside diameter of the compound idler gears.

Large Idler Gear Inside Diameter		
mm		in
73.50	MIN	2.894
73.53	MAX	2.895

Small Idler Gear Outside Diameter (Diameter that fits in large gear)		
mm		in
73.44	MIN	2.891
73.47	MAX	2.892

Check parts for damage.

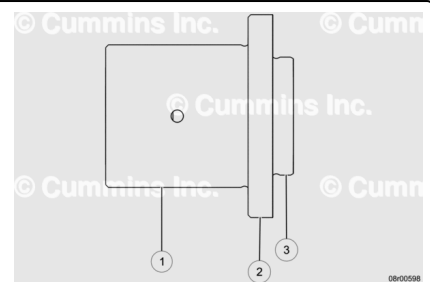
Measure the inside diameter of the small idler bushing.

Small Idler Bushing Inside Diameter		
mm		in
52	MIN	2.047
52.06	MAX	2.049

Note : The bushing in the gear is bored after installation. If machining capability is not available, replace the bushing and gear as an assembly.

Check parts for damage.

Measure the outer diameters of the idler shaft.



Idler Shaft Outer Diameters

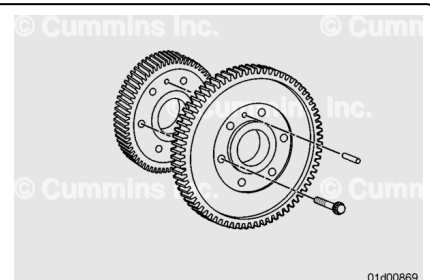
	mm		in
1	51.96	MIN	2.044
	51.973	MAX	2.046
2	74.02	MIN	2.914
	73.8	MAX	2.905
3	42.988	MIN	1.692
	43.012	MAX	1.693

Install

If the shaft is out of specification, discard and replace with a new idler shaft.

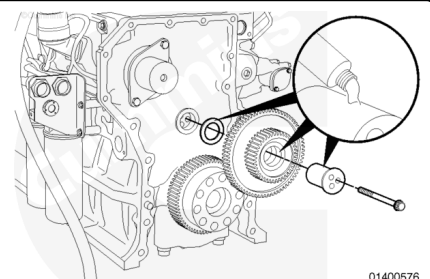
Align the large gear and small gear using a dowel pin.

Torque Value: 110 n•m [80 ft-lb]



⚠ CAUTION ⚠

The chamfer in the thrust washers must be positioned toward the gear. Improper installation of the thrust washer can cause engine damage.



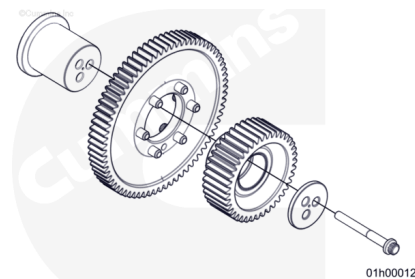
For CKEC engines built before ESN 00327150, the timing marks on the camshaft gear **must** be visible when the gear is installed.

Use Lubriplate™ 105, or equivalent, to lubricate the gear bushing, shaft, and the retainer plate.

After lubricating, align the idler shaft, gear, and retainer plate before installing onto the block.

⚠ CAUTION ⚠

The chamfer in the thrust washers must be positioned toward the gear. Improper installation of the thrust washer can cause engine damage.



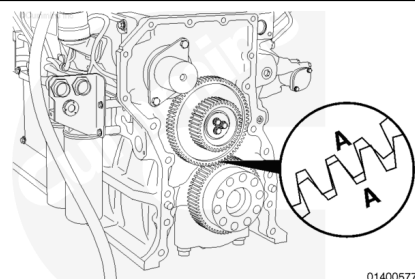
For CKEC engines built after ESN 00327150, and PHP engines built after ESN 85000848, the timing marks on the camshaft gear **must** be visible when the gear is installed.

Use Lubriplate™ 105, or equivalent, to lubricate the gear bushing, shaft, and the retainer plate.

After lubricating, align the idler shaft, gear, and retainer plate before installing onto the block.

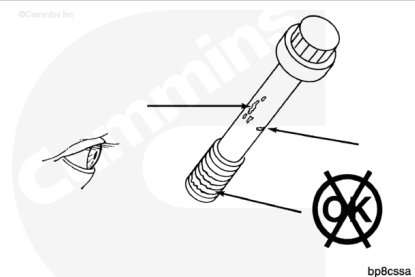
The camshaft idler gear is the **only** idler gear that has timing marks.

Align the "A" on the larger idler gear with the "A" on the crankshaft gear.



⚠ CAUTION ⚠

Check the punch mark on the mounting capscrew head. If seven marks are punched, replace the capscrew with a new capscrew. Failure to replace the capscrew could result in breakage.



Capscrew Inspection Criteria

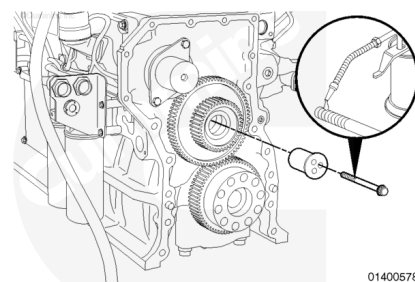
- Inspect the capscrews for damage.

The capscrews **must** be replaced if:

- Seven punch marks are visible on the head of the capscrew.
- The threads are damaged.
- Rust or corrosion has caused pitting in the body.
- The body is nicked, galled, bent, or stretched.
- Fretting is found under the capscrew head.

⚠ CAUTION ⚠

The outer face of the idler gear shaft is stamped with the word "UP". This stamp must be positioned toward the cylinder heads. Improper installation of the idler gear shaft can cause engine damage.



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Use clean engine oil to lubricate the capscrews.

Use the capscrew to pull the shaft into the bore.

Use the following steps to tighten the capscrews in the sequence shown.

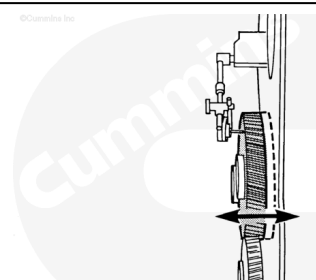
Torque Value:

1. 45 n•m [35 ft-lb]
2. 60 n•m [44 ft-lb]
3. 90 degrees

After tightening capscrew, apply a punch mark to the head of the capscrew.

Use a dial indicator to measure the idler gear end clearance.

Camshaft Idler Gear End Clearance		
mm		in
0.04	MIN	0.002
0.18	MAX	0.007



if4001a

If the clearance is **not** within specifications, check for foreign material between the parts, or check for proper location of the thrust washers. Oversize thrust washers are available.

Camshaft Idler Gear Lash		
mm		in
0.145	MIN	0.006
0.380	MAX	0.0150

Finishing Steps

- Install the camshaft gear. Refer to Procedure 001-012 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-012.html>).
- Install the flywheel and flywheel housing. Refer to Procedure 001-008 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-008.html>).

- Install the oil pan and related components. Refer to Procedure 007-025 in Section 7 (/qs3/pubsys2/xml/en/procedures/89/89-007-025.html).

Last Modified: 08-Feb-2023
